

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit

StudentGradeTestAAA.Java:

```
package org.example;

import org.junit.After;
import org.junit.Before;
import org.junit.Test;

import static org.junit.Assert.assertEquals;

public class StudentGradeAAATest {

    private StudentGrade grade;

    @Before
    public void setUp() {
        System.out.println("Setting up...");
        grade = new StudentGrade();
    }

    @After
    public void tearDown() {
        System.out.println("Test Finished.");
    }

    @Test
    public void testGradeA() {

        // Arrange
        int marks = 95;

        // Act
        String result = grade.getGrade(marks);

        // Assert
        assertEquals("A", result);
    }

    @Test
    public void testGradeB() {

        // Arrange
        int marks = 80;

        // Act
        String result = grade.getGrade(marks);
```

```
        // Assert
        assertEquals("B", result);
    }

    @Test
    public void testGradeF() {

        // Arrange
        int marks = 20;

        // Act
        String result = grade.getGrade(marks);

        // Assert
        assertEquals("F", result);
    }
}
```

OUTPUT:

